

Phased fuel transitions for decarbonizing the Asia–Europe Green Shipping Corridor

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International shipping stands at a pivotal juncture in its decarbonization efforts, yet a comprehensive assessment of viable fuel transition pathways remains scarce. Here we develop a dynamic life cycle assessment framework integrated with system dynamics to evaluate 13 alternative marine fuels (for example, ammonia, methanol, hydrogen and soybean-based biodiesel) for the Asia–Europe Green Shipping Corridor through 2050. The framework incorporates climate impacts, environmental sustainability, energy efficiency and economic feasibility. Under three scenarios (Business as Usual, Most Economical and Maximum Carbon Reduction), this study accounts for key dynamic parameters, including technology learning curves. Results indicate a phased transition: soybean-based biodiesel offers a near-term solution, while electrofuels dominate after 2040. The International Maritime Organization’s Net-Zero Framework progressively narrows regional cost–emission gaps, whereas the European Union’s FuelEU Maritime regulation introduces a cost premium of up to 7.6%. These results indicate that policy design and regulatory stringency substantially shape fuel transition trajectories for maritime decarbonization.

International maritime shipping is a fundamental sector of global trade, facilitating approximately 90% of the world’s merchandise movement¹. Yet, it remains a substantial emitter of greenhouse gas (GHG) emissions, contributing roughly 3% of the global emissions². Despite mounting regulatory pressures, the sector’s decarbonization progress remains limited, with over 99% of its energy still derived from fossil fuels³, highlighting a stark disconnect between ambitious net-zero targets and the current pace of transition. In response, the International Maritime Organization (IMO) has progressively strengthened its emissions reduction targets. Its 2018 Initial GHG Strategy aimed to halve annual emissions by 2050⁴, which was updated in 2023 through a Revised Strategy committing to achieve net-zero GHG emissions ‘by or around 2050’. The revised strategy also sets stricter interim targets: a 40% reduction by 2030 and 70–80% by 2040 (relative to a 2008 baseline)⁵. To realize these ambitions, the IMO has proposed a Net-Zero Framework

prioritizing the expansion of biofuels and other alternative fuels within the maritime energy portfolio⁶. Although the formal adoption of this framework has been delayed⁷, global momentum towards achieving net-zero emissions continues. This is exemplified by regional initiatives such as the European Union (EU)’s FuelEU Maritime regulation, set to launch in 2025⁸, which aims to accelerate the transition and serve as a testing ground for decarbonization pathways.

Navigating this complex, multipolicy landscape to identify viable transition pathways under deep uncertainty is therefore a critical challenge. In this context, Green Shipping Corridors, defined as specific maritime routes where stakeholders collaborate to adopt zero-emission fuels ahead of broader industry timelines, have emerged as a strategic focal point⁹. Focusing on a central corridor, such as the Asia–Europe Green Shipping Corridor, can therefore reduce investment risks and develop scalable industry-wide decarbonization solutions.

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In practice, deep decarbonization faces multiple hurdles. Existing ship technologies, such as onboard emission capture systems, can deliver only moderate emission reductions¹⁰. Port infrastructure upgrades also require extended lead times and are often misaligned with regulatory timeframes¹¹. In addition, electric ships are not yet viable for long-distance routes¹². Consequently, the large-scale deployment of low-carbon alternative fuels, such as hydrogen, ammonia, methanol and biofuels¹³, has increasingly been recognized as the most feasible pathway to meet the targets of the FuelEU Maritime regulation and the IMO Net-Zero Framework.

However, reliably evaluating this critical fuel transition under real-world conditions is complex. Shipping regulations and technologies are shifting, as illustrated by the delay in approving the IMO Net-Zero Framework, creating persistent uncertainty. Current analytical tools, however, are not equipped to fully capture this complexity. Traditional assessment frameworks, primarily based on static life cycle assessment (LCA)^{14,15} or limited dynamic projections¹⁶, offer only snapshot comparisons that fail to account for the critical feedback loops through which technical learning curves, fuel adoption and infrastructure deployment interact and co-evolve, the factors that ultimately determine the long-term feasibility of transition pathways. Furthermore, most existing pathway analyses treat regulations as static, external factors rather than as endogenous variables that interact dynamically with technological and economic factors^{17,18}.

To address this gap, we developed a dynamic LCA framework that integrates system dynamics (SD), moving beyond static snapshots to capture the evolving, multidimensional performance of alternative marine fuels through 2050. It assesses four dimensions: climate impacts, environmental sustainability, energy efficiency and economic feasibility (termed Climate-3E). The framework is applied to a critical case study: the Asia–Europe Green Shipping Corridor, selected for its high trade volume and potential exposure to dual regulatory pressures from the IMO Net-Zero Framework and the FuelEU Maritime regulation. The analysis encompasses 13 alternative fuels, benchmarked against heavy fuel oil (HFO) and marine diesel oil (MDO), including biodiesel, denoted as soybean-based fatty acid methyl ester (FAME); methanol variants: coal-based (Coal-MeOH), natural gas-based (NG-MeOH), electro-MeOH (e-MeOH) and bio-derived (bio-MeOH); liquefied natural gas variants: conventional (Conv-LNG), electro-LNG (e-LNG) and bio-derived (bio-LNG); liquid hydrogen variants: conventional (Conv-LH₂), electro-LH₂ (e-LH₂) and bio-derived (bio-LH₂); liquid ammonia variants: conventional (Conv-LNH₃) and electro-LNH₃ (e-LNH₃).

This study addresses a central question for the maritime low-carbon transition: which fuel transition pathways can most effectively decarbonize global shipping under the evolving IMO Net-Zero Framework and the FuelEU Maritime regulation? The proposed dynamic modelling framework reveals how policy timing and regulatory stringency may shape future mitigation outcomes, clarifying the conditions under which alternative fuels gain or lose competitiveness. While the modelling framework cannot encompass all factors influencing decision-making, it provides forward-looking, scenario-based projections of fuel competitiveness and transition trajectories, thereby empowering shipowners, policymakers and investors to make informed decisions. By illuminating the dynamic interplay of policy, technology and fuel competitiveness, this study supports confident, timely commitments to the most robust and scalable pathways for a climate-resilient maritime future.

Results

Current status and prospects of alternative marine fuels

A comprehensive LCA is conducted to evaluate the decarbonization potential of marine fuels by quantifying their Climate-3E performance (climate impacts, environmental sustainability, energy efficiency and economic feasibility) relative to baseline HFO and MDO. Across 1,300 scenarios (Supplementary Fig. 3) reflecting diverse production

pathways and operational conditions, the assessment reveals a spectrum of performance for alternative fuels, as detailed in Fig. 1.

Figure 1a shows that hydrogen-based marine fuels, particularly electro- and bio-LH₂, achieve near-zero life cycle GHG emissions in most scenarios. Conv-LH₂ also delivers substantial GHG emission reduction compared with HFO and MDO. However, the low energy density of hydrogen fuels limits their practical use, often requiring 30–70% blending with diesel on long voyages, diminishing overall climate benefits^{19,20}. Bio-LNG emerges as a viable transitional fuel, offering substantial emission reductions due to biogenic carbon uptake²¹. By contrast, Conv-LNG exhibits higher life cycle emissions than HFO and MDO due to methane leakage and the energy-intensive liquefaction process²². E-LNG performs between bio-LNG and HFO under current energy structures. All LNG-based pathways retain lower long-term emissions, underscoring their continued relevance for the low-carbon transition.

Conversely, methanol- and ammonia-based fuels, including their electrofuel (e-fuel) variants, generally exhibit higher life cycle emissions than HFO and MDO, because their low energy density increases fuel consumption per distance travelled, and fossil-derived variants (Coal-MeOH, NG-MeOH and Conv-LNH₃) rely on carbon-intensive synthesis²³. While bio-MeOH offers meaningful emission reduction, the climate performance of e-MeOH and e-LNH₃ remains strongly constrained by the carbon intensity of the electricity mix. Under current fossil-dominated grids, their carbon footprints may exceed those of HFO and MDO.

Figure 1b–d compares the carbon intensity indicator (CII) (a regulatory metric established by the IMO to quantify operational carbon efficiency), energy efficiency and economic viability^{24,25}. HFO and MDO exhibit high CII values due to their high tank-to-wake emissions resulting from the combustion of carbon-intensive hydrocarbons^{26,27}. Biofuels, by contrast, demonstrate balanced advantages. For example, bio-MeOH, bio-LNG and bio-LH₂ may achieve emission reductions of 41.5%, 47.4% and 31.1%, respectively, compared with HFO and offer more favourable energy and economic profiles than most e-fuel alternatives. However, scalability and slightly higher costs remain barriers, and e-fuels provide deep emission reductions when powered by renewable electricity²⁸, but their production remains highly energy-intensive²⁹. The electrolysis process for hydrogen production, a foundational stage for all these e-fuels, entails substantial energy conversion losses³⁰, and both carbon capture and electrolysis technologies exhibit high costs^{28,31,32}. As a result, their current market competitiveness remains low, with only 11.1–35.6% of HFO.

Technological advancements are expected to shift this landscape. Rapid investment in e-fuels and direct air capture of atmospheric CO₂ is expanding production potential^{33,34}. By 2050, the Climate-3E performance of e-LNG, bio-MeOH, e-LNH₃ and e-MeOH is projected to surpass that of HFO and MDO. Meanwhile, the results also reveal policy-dependent divergence. While the Climate-3E performance under both policies is closely aligned, the FuelEU Maritime regulation provides more substantial economic incentives for fuel transition compared with the IMO Net-Zero Framework (Supplementary Figs. 16–18).

Sensitivity analysis shows that e-fuels (e-MeOH and e-LNG) exhibit greater stability against carbon price fluctuations (±20%) than fossil-derived and biomass-derived fuels, which show greater volatility in emissions and costs (Supplementary Fig. 10). Life cycle impact assessment results indicate that biofuels offer cobenefits such as lower acidification and eutrophication potentials. At the same time, certain e-fuels and ammonia pathways entail higher human and ecotoxicity burdens under the current energy system (Supplementary Fig. 11). A breakdown of life cycle costs confirms that fuel costs dominate, with e-MeOH the most expensive and Conv-LNG among the most cost-competitive alternatives when ship modifications and maintenance are considered (Supplementary Fig. 12). These findings underscore that, while e-fuels promise long-term decarbonization and price resilience, biofuels present a balanced near-term profile with favourable environmental cobenefits, albeit with scalability constraints.

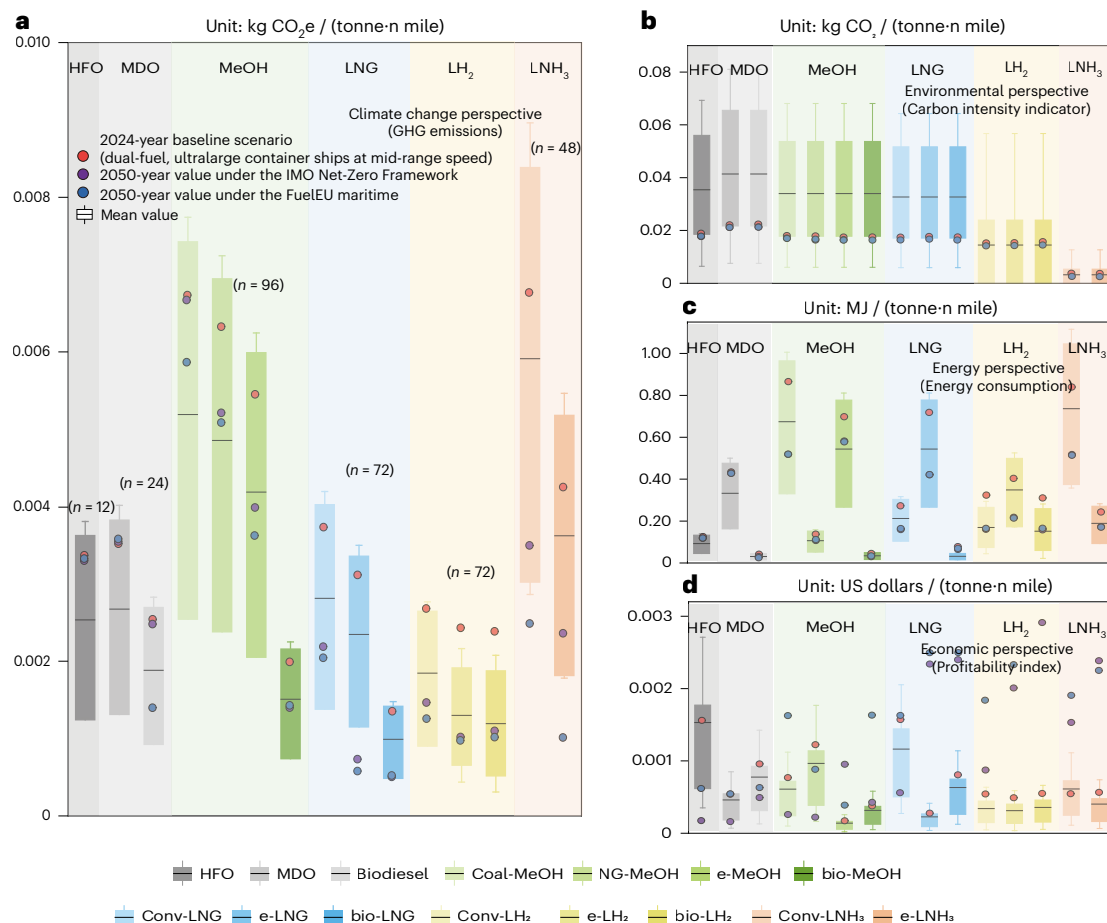


Fig. 1 | Life cycle assessment of marine fuels under the IMO Net-Zero Framework and FuelEU Maritime. **a–d**, Life cycle GHG emissions (**a**), carbon intensity indicator (**b**), energy consumption (**c**) and profitability index (**d**) of marine fuels in the context of the evolving IMO Net-Zero Framework and the FuelEU Maritime regulation. Each fuel pathway is modelled in a self-contained scenario in which its market adoption competes dynamically with HFO based on economics, policies and infrastructure within our framework, which integrates LCA for the baseline and SD projections for 2050. Box plots show the distribution of life cycle values across different vessel capacities, operational

scenarios and electricity-generation structures under current (2024) conditions (Supplementary Fig. 3 and Supplementary Table 6). Sample size: $n = 324$ independent scenarios; each scenario is a distinct deterministic LCA simulation. Box plot: the centre line indicates the median; the lower and upper box bounds mark the 25th and 75th percentiles; the whiskers extend to the minimum and maximum values. Dots: the baseline scenario dot ($n = 1$) and the two 2050 policy projection dots (from the SD model). The unit 'tonne-n mile' represents carrying one metric tonne of cargo over one nautical mile.

How the electricity mix shapes e-fuel performance

The performance of e-fuels is powerfully shaped by the carbon intensity of the electricity used in their production. To evaluate this influence in a strategic decarbonization context, this study focuses on the Asia–Europe Green Shipping Corridor, which has high trade volumes and is likely to face dual regulatory pressures from the IMO Net-Zero Framework and the EU's FuelEU Maritime regulation. Figure 2 summarizes the installed renewable power generation capacity along this corridor³⁵, highlighting regional disparities that underlie fuel competitiveness.

East Asia has the highest renewable power capacity (649 GW), while Western Europe leads in the share of renewables within its electricity mix. To assess the impact, fuel performance is compared under the current electricity mix and renewable power scenario using the Climate-3E framework. Results for the EU are shown in Fig. 3 (results for other regions along the Asia–Europe Green Shipping Corridor shown are in Supplementary Fig. 15).

From a climate perspective (Fig. 3a), renewable electricity is pivotal for e-fuels to achieve deep-life cycle GHG cuts. Conventional and biomass-derived fuels, such as Coal-MeOH, NG-MeOH, Conv-LNG and bio-variants, exhibit limited sensitivity to the electricity mix, whereas e-fuels show pronounced variation. Under the renewable electricity

scenario, e-LNG and e-LNH₃ achieve low emissions, while e-MeOH remains carbon-intensive due to process-related N₂O and CH₄ emissions during synthesis³⁶.

Environmental performance (Fig. 3b) further illustrates the influence³⁷. The carbon intensity of e-fuels declines substantially as renewable electricity penetration increases. Biofuels remain stable in carbon intensity because they rely little on electricity. Under the renewable scenario, Conv-LNH₃ achieves a carbon intensity of 0.002 kg of CO₂ per tonne-nautical mile, below the regulatory benchmark, and both e-MeOH and e-LNG can attain negative carbon intensity.

Energy performance (Fig. 3c) shows that renewable power has a limited impact on conventional and biomass-derived fuels, whose life cycle energy consumption depends on the feedstock and process technology. By contrast, renewable power substantially reduces the energy intensity of e-fuel production³³. For example, the life cycle energy consumption of renewable-powered e-MeOH, e-LNG and e-LNH₃ is reduced by 57.2%, 56.8% and 41.5%, respectively. This improvement stems from the higher efficiency of renewable electricity systems relative to fossil-derived thermal generation. Economic outcomes (Fig. 3d) are less responsive to changes in the electricity mix. However, the declining cost of renewable energy enhances the long-term competitiveness of e-fuels^{38,39}. Continued advances in renewable technologies

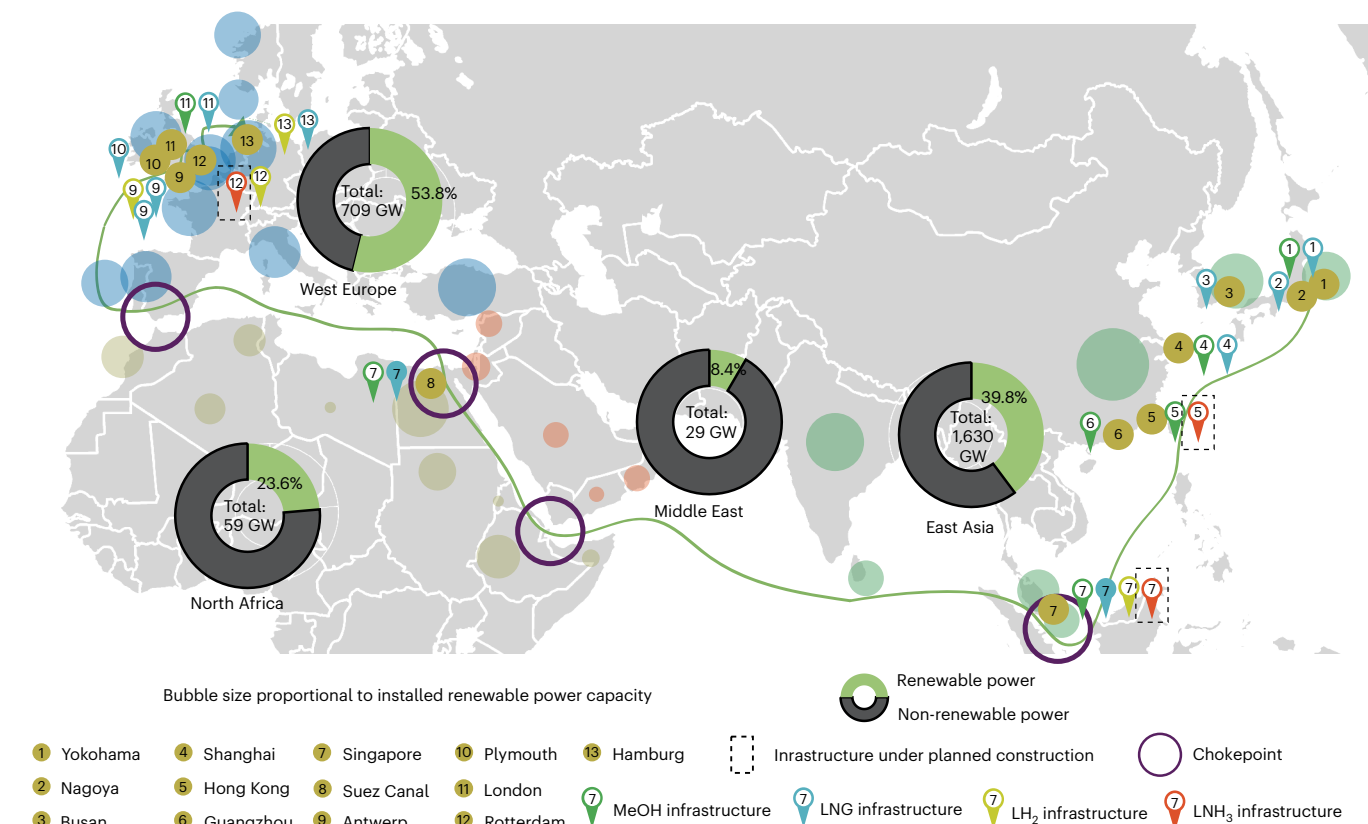


Fig. 2 | Refuelling ports and renewable energy profile along the Asia–Europe Green Shipping Corridor. This figure presents the installed renewable power capacity of countries and regions along the Asia–Europe Green Shipping

Corridor, accounting for the availability of renewable fuels, the electricity mix and regional development. Basemap data from Natural Earth (<https://www.naturalearthdata.com>).

and the scaling of e-fuel production are expected to strengthen their economic viability further.

Annual costs and emissions under dual regulatory scenarios

The transition to low-carbon marine fuels is a long-term, dynamic, path-dependent process. An analysis is therefore essential to capture total emissions over the transition period, avoiding misinterpretation from short-term fluctuations. Guided by this perspective, transition outcomes across key regions along the Asia–Europe Green Shipping Corridor—from Japan, through China (East Asia), Malaysia (South-east Asia), the United Arab Emirates (Middle East), Egypt (Africa), to Europe—are compared under different policy scenarios, as simulated in Fig. 4.

Figure 4 compares annual carbon emissions under three scenarios: Business as Usual (BAU), the Most Economical (ME) and the Maximum Carbon Reduction (MCR) for the IMO Net-Zero Framework-only, the FuelEU Maritime-only and two policies combined. The IMO Net-Zero Framework has considerable potential to align economically viable and low-emission pathways across regions. By contrast, the FuelEU Maritime could lead to greater divergence between these objectives. This contrast reflects fundamental differences in policy design. The IMO Net-Zero Framework combines stringent annual carbon-intensity reductions with a two-tier carbon-emission penalty system (US\$100 per tonne CO₂ equivalent (CO₂e) for tier 1 and US\$380 per tonne CO₂e for tier 2)⁴⁰. By contrast, the EU applies more gradual carbon-intensity targets, but imposes a higher carbon-emission penalty of US\$730 per tonne CO₂e⁴¹, nearly twice the IMO tier 2 rate. Because Conv-LNG can meet EU compliance standards for an extended period and provides high economic efficiency, shipowners are incentivized to continue using Conv-LNG as an alternative fuel, suppressing the growth of e-fuels. Conversely, the IMO's stricter trajectory discourages long-term

reliance on transitional fossil fuel options, accelerating the competitiveness and uptake of e-fuels and biofuels.

Regional outcomes reflect these regulatory dynamics. Under the IMO Net-Zero Framework, Japan's ME and MCR pathways yield similar annual carbon emissions, indicating strong convergence between economic and environmental objectives. Compared with the IMO Net-Zero Framework-only scenario, the two policies combined scenario yields only a small difference in annual carbon emissions. This is because, under the combined regime, the overall carbon emission trajectory converges towards the more stringent IMO targets, restricting prolonged use of transitional fuels and thus preventing a substantial divergence between the ME and MCR pathways. However, under the FuelEU Maritime-only scenario, a clear gap exists between economic and environmental objectives within China, which widens in regions with electricity systems dominated by renewable energy. Conversely, it can narrow further in the carbon-intensive regions. Because Conv-LNG only ensures near-term compliance at low cost, it pulls carbon-intensive regions towards similar ME and MCR emissions trajectories, thereby accelerating the transition to cleaner fuels. These findings demonstrate that the FuelEU Maritime's lenient carbon-intensity targets may inadvertently prolong the use of conventional fuels, even imposing higher penalties, particularly in regions with high renewable penetration. By contrast, the IMO Net-Zero Framework's annually tightened emission-mitigation targets can promote both emission reductions and economic feasibility across regions without imposing disproportionately higher costs, thereby enabling more consistent global progress towards maritime decarbonization.

Technological pathways for marine fuel transitions

The adoption of alternative marine fuels involves distinct technological strategies to achieve decarbonization (Supplementary Fig. 8). Current

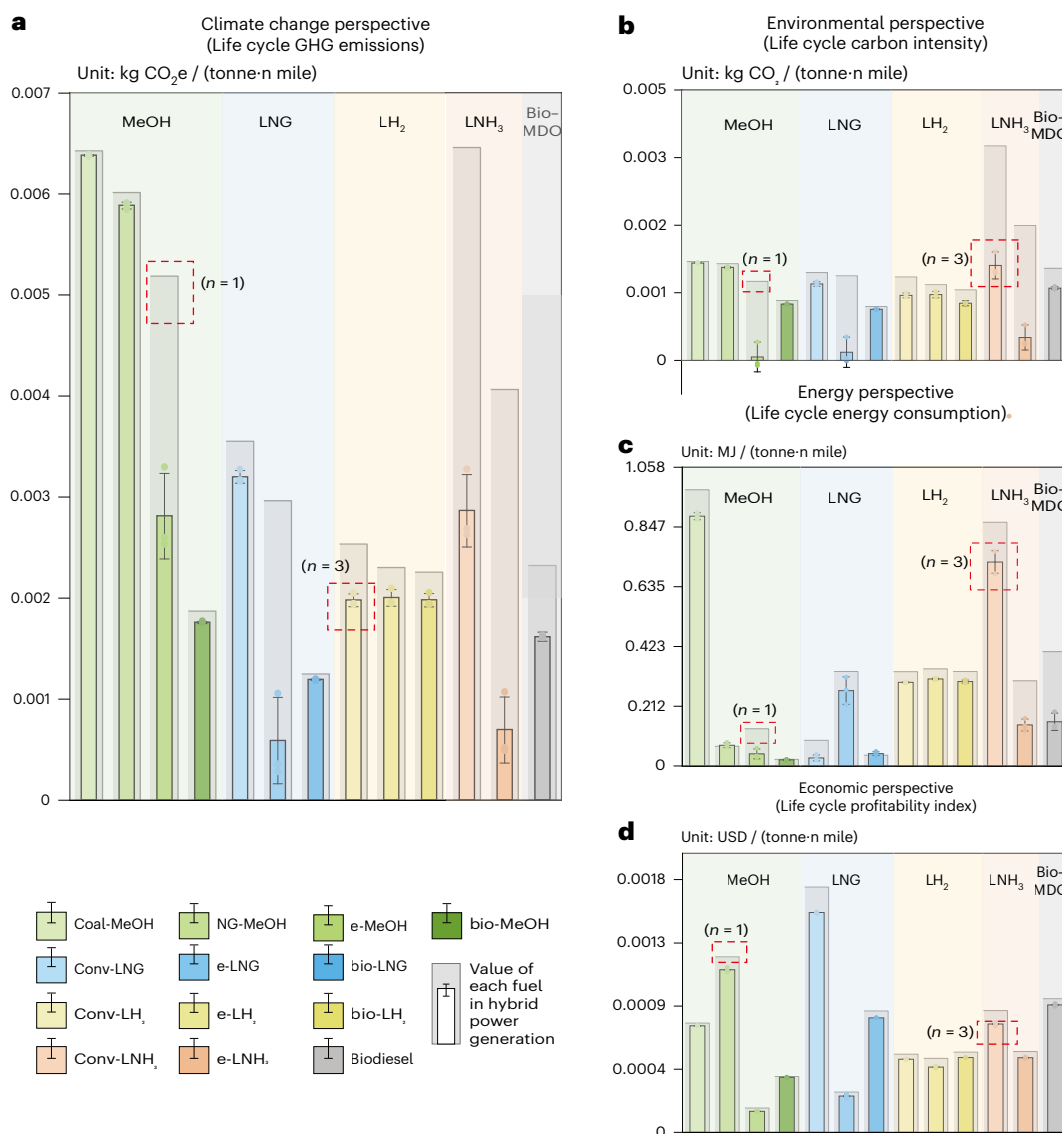


Fig. 3 | The Climate-3E performance of alternative marine fuels under different electricity mixes. a–d, Life cycle GHG emissions (a), life cycle carbon intensity (b), life cycle energy consumption (c) and life cycle profitability index (d) of marine fuels from Climate-3E perspectives. Results are derived from LCA analyses of ultralarge containerships at mid-range speeds. The fuel performance value for the EU current electricity mix is derived from a single deterministic

scenario ($n = 1$). The coloured bars representing the renewable electricity mix (wind, solar and hydro) show the mean of three independent simulations ($n = 3$, one per renewable source); the error bars indicate the minimum–maximum range, and the overlaid dots display the individual values for wind, solar and hydropower. Each data point corresponds to an independent scenario.

methods include drop-in fuel blending, retrofitting conventional engines and deploying new dual-fuel engines capable of operating on MeOH, LH₂ or LNH₃ (refs. 42,43). Figure 5 presents projected fuel strategies from 2024 to 2050, along with their associated carbon intensity and relative costs, under two regulatory scenarios: the FuelEU Maritime-only scenario and the two policies combined scenario, revealing distinct technological trajectories. Drop-in biodiesel blending with MDO offers meaningful near-term emission reductions, lowering carbon intensity to 0.047 kg of CO₂e per megajoule. However, this pathway remains economically constrained and fails to comply with the IMO targets beyond 2035, serving only as a short-term transitional measure.

Newbuild dual-fuel engines exhibit more complex and policy-dependent outcomes. While e-MeOH delivers low life cycle carbon emissions, its economic performance varies considerably across policies. It remains relatively affordable under the FuelEU Maritime-only scenario, but becomes more expensive under the two policies combined scenario. Both e-LNG and bio-LNG show strong

decarbonization potential, whereas Conv-LH₂ and Conv-LNH₃ remain cost-effective but carbon-intensive. Their electro-counterparts face substantial technical hurdles: hydrogen still requires substantial diesel blending, and ammonia relies on partial diesel use; both incur high compliance costs, especially under the IMO Net-Zero Framework-only scenario (Supplementary Fig. 19). This scenario supports e-LNG and bio-LNG, while the FuelEU Maritime-only scenario supports e-MeOH and e-LNG. Under the combined policy scenario, however, high carbon compliance costs raise the cost of conventional fuels such as HFO, thereby enhancing the economic viability of other low-carbon alternatives, notably e-LH₂ and e-LNH₃. These comparisons illustrate that the selection of a technological pathway is not determined solely by inherent fuel qualities but is influenced by the policy framework.

Discussion

Achieving deep decarbonization in the shipping industry requires diverse, multistage transition pathways that account for the distinct

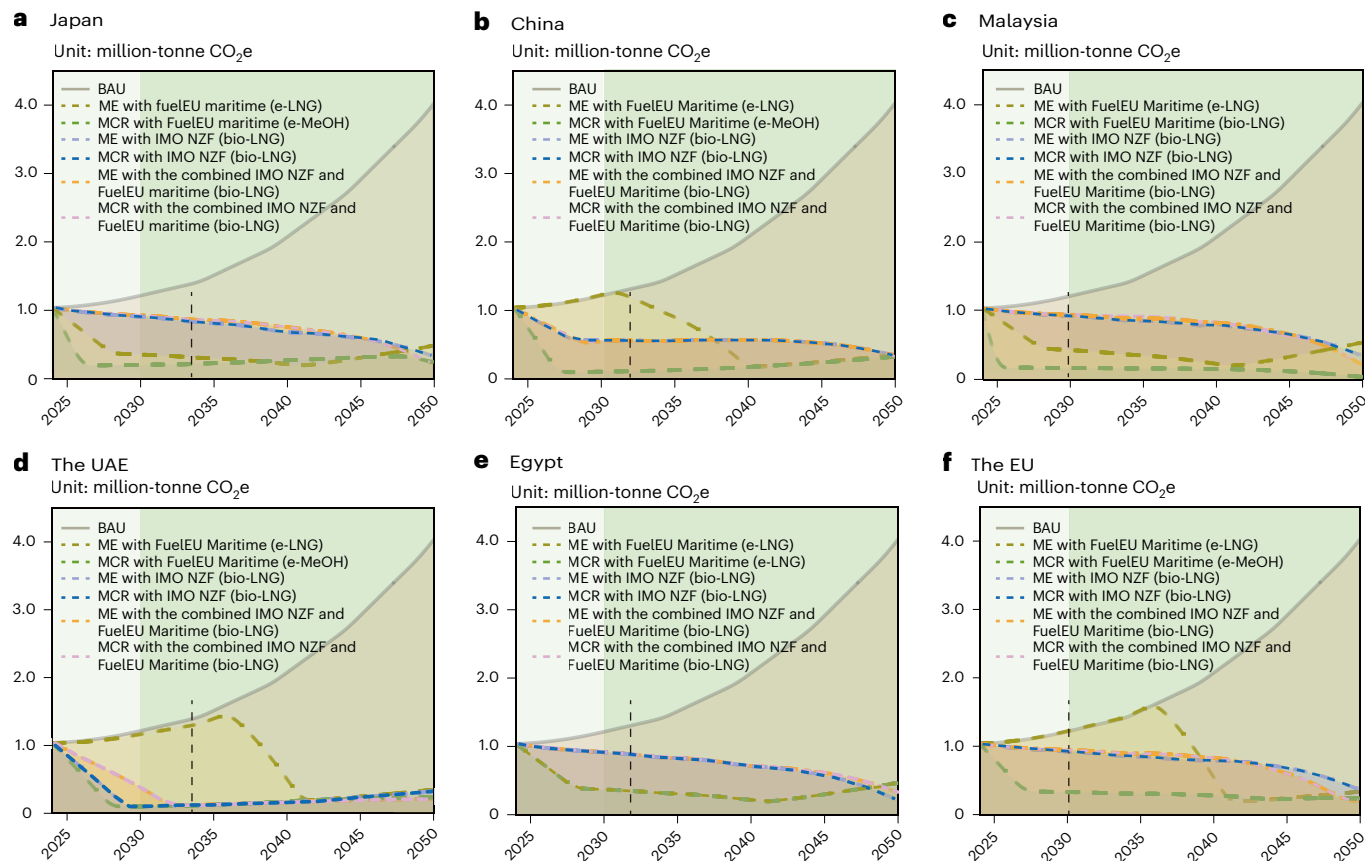


Fig. 4 | Simulated annual carbon emission differences across six regions along the Asia–Europe Green Shipping Corridor under different policy scenarios (2024–2050). **a–f**, Simulated annual carbon emissions of Japan (**a**), China (**b**), Malaysia (**c**), the UAE (**d**), Egypt (**e**) and the EU (**f**). The analysis models a round-trip voyage cycle originating from and returning to a port within each specified region and assumes that the vessels operate on a consistent fuel blend.

NZF refers to the Net-Zero Framework. BAU: continuous use of HFO without policy intervention; ME: a fuel transition strategy balancing economics and compliance within the Climate-3E framework; MCR: a pathway focused on minimizing emissions by adopting the most technically feasible low-carbon fuels. Projections incorporate the mechanisms of the evolving IMO Net-Zero Framework and the FuelEU Maritime regulation.

production routes and performance profiles of alternative marine fuels. From a climate perspective, biodiesel, hydrogen and bio-LNG offer substantial potential to reduce life cycle GHG emissions. By contrast, fossil-derived alternatives provide limited climate benefits due to their energy-intensive production processes. Economically, Conv-LNG already delivers notable cost savings over HFO and MDO, while other biofuels such as bio-MeOH and bio-LNG remain moderately more expensive, and e-fuels still carry a substantial price premium. Overall, biofuels represent the most balanced near-term options, offering a viable combination of economic competitiveness and tangible emission reductions.

A critical perspective emerges when considering the inherent uncertainties in these comparisons. The probability distributions of the Climate-3E metrics from the Monte Carlo analysis (Supplementary Fig. 9) quantify this uncertainty, revealing a spectrum of possible outcomes. This underscores that the real-world viability of any pathway cannot be judged solely by its average performance. For instance, the wide distribution of the CII highlights a tangible compliance risk for several fuels under variable operating conditions. This performance uncertainty translates directly into economic risks. The sensitivity of fossil-derived and biomass-derived pathways to carbon price fluctuations (Supplementary Fig. 10) exposes early adopters to substantial financial and regulatory volatility. Conversely, the price resilience of e-fuels offers a long-term hedge. Therefore, the fuel transition is not merely a cost–emission trade-off but a strategic choice between accepting short-term volatility or investing in long-term predictability. This uncertainty-aware perspective calls for policies

that not only penalize emissions but also actively derisk investments in stable, deep-decarbonization technologies.

Regional variation in electricity mixes further shapes the suitability of fuels along green shipping corridors. In regions with a high share of renewable electricity, e-fuels achieve notable reductions in GHG emissions and carbon intensity, demonstrating strong long-term decarbonization potential. Conversely, where fossil fuels dominate the electricity mix, the carbon intensity of e-fuels remains high, severely limiting their environmental value. The prospects of biofuels, meanwhile, depend on the availability of biomass feedstock (Supplementary Information section IV). Given the decisive role of clean electricity in determining life cycle emissions, addressing regional differences in power generation is essential to designing effective transition pathways. Efforts should therefore focus on deploying location-appropriate fuels, expanding renewable capacity in fossil-dependent regions and preparing the infrastructure necessary for future large-scale e-fuel adoption.

Maritime decarbonization is further shaped by regulatory architecture. The FuelEU Maritime regulation advances emissions reductions through a penalty system and moderate carbon-intensity targets, enabling meaningful reductions in specific contexts. However, its effectiveness varies by region. In areas with cleaner power grids, Conv-LNG is likely to retain a long-term compliance advantage due to its lower costs, potentially slowing the adoption of higher-cost, lower-emission alternatives. Conversely, in regions with carbon-intensive grids, transition fuels cannot meet long-term compliance requirements, prompting a shift to alternative fuels to avoid steep penalties, thereby

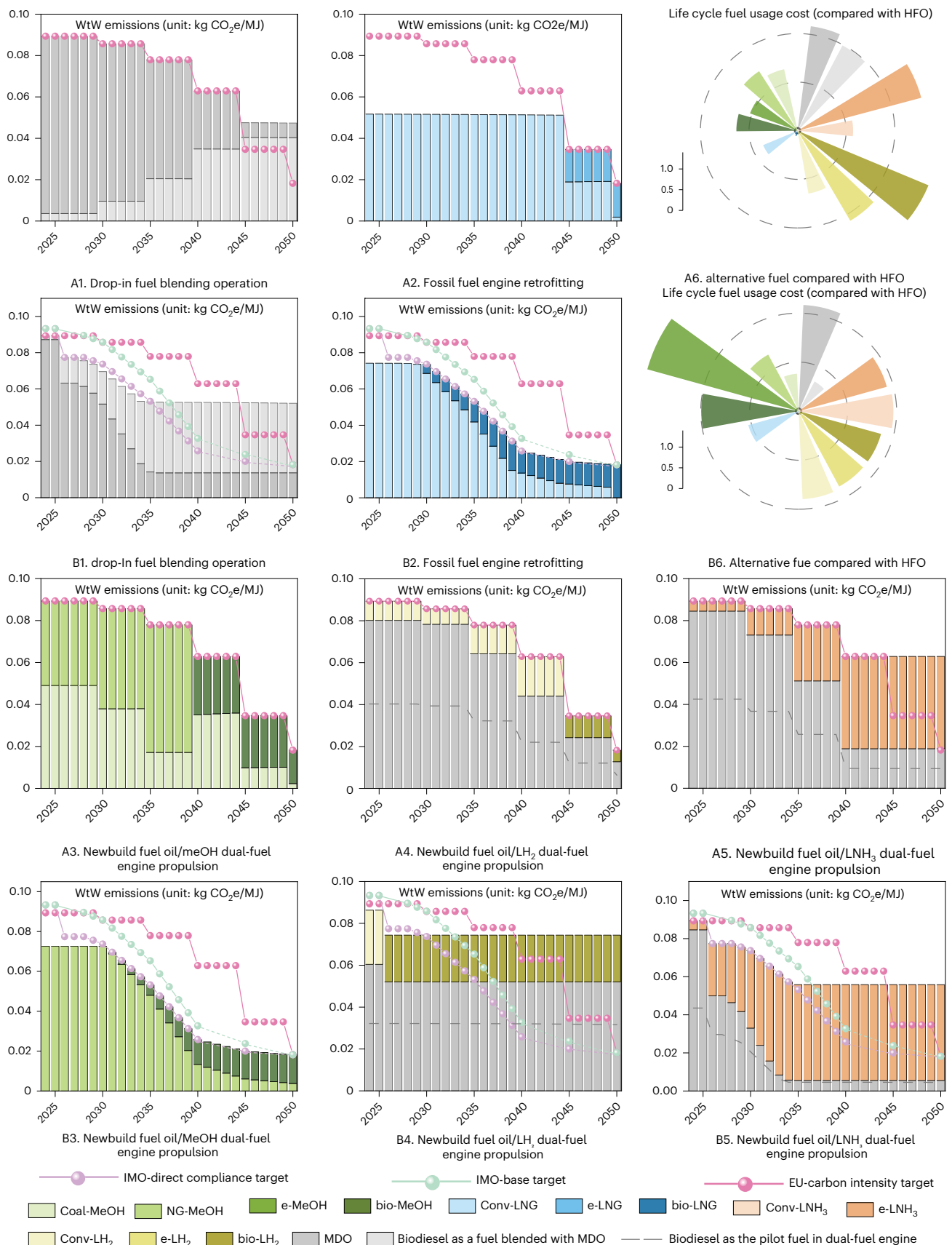


Fig. 5 | Future fuel strategies and relative fuel expenditure levels in 2050.

Results from the integrated LCA–SD framework under the FuelEU Maritime-only scenario (labelled ‘A’) and the two policies combined scenario (labelled ‘B’). The charts project the evolution of fuel strategies to 2050 and their associated carbon

intensity. The inset table summarizes the relative fuel cost multiplier compared with HFO for each strategy in 2050. All results are reported on a well-to-wheel (WtW) basis. The simulation reveals distinct differences in fuel competitiveness shaped by the policy structure.

accelerating the transition to low-carbon fuels and amplifying shipping decarbonization outcomes.

In comparison, the IMO Net-Zero Framework uses a two-tier carbon-pricing mechanism coupled with strict, annually tightening carbon-intensity targets. This design steadily narrows the gap between the most economical pathways and those with the strongest emission reductions, effectively balancing environmental ambition with economic feasibility. Consequently, the IMO Net-Zero Framework promotes greater convergence in decarbonization pathways across regions with heterogeneous electricity mixes, demonstrating broader applicability. It provides a more globally balanced approach, integrating environmental rigour with economic practicality and supporting scalable transition strategies. When the FuelEU Maritime regulation is already in place, layering the IMO Net-Zero Framework on top can further reduce the cost differential between alternative and conventional marine fuels. As a comprehensive global benchmark, the IMO Net-Zero Framework is better positioned to deliver consistent, equitable and scalable progress towards decarbonizing international shipping.

Methods

The integrated LCA–SD framework

To dynamically evaluate the multidimensional performance and transition pathways of alternative marine fuels, we developed an integrated framework combining LCA and SD (Supplementary Figs. 1 and 2). Among various scenario-based projection approaches, SD modelling is particularly well suited due to its ability to transparently represent the interactions among system elements, making results more interpretable and easier to validate. In addition, SD can simulate dynamic behaviour and account for feedback effects over time⁴⁴. This method has been widely applied in the energy sector to project technology diffusion and policy impacts. Its main advantage lies in its capacity to integrate multiple subsystems, aligning with the multistage, interactive nature of LCA. The SD model was constructed and simulated in Vensim PLE (version 10.4.0), in which the model's causal structure, including key feedback loops related to policy compliance, technology costs, fuel adoption decisions and infrastructure investment, was formally represented as a stock-and-flow diagram (Supplementary Figs. 5 and 6). A comprehensive review of prior static and dynamic LCA studies, along with a systematic comparison of alternative methodologies, is provided in Supplementary Tables 1–3.

The assessment follows an eight-step process organized into two interlinked phases. The first phase involves a static LCA to establish a robust, multidimensional baseline of fuel performance. We began by defining the scope of the analysis, focusing on the Asia–Europe Green Shipping Corridor, selecting 13 alternative marine fuels. Key operational parameters, including representative containership sizes, were specified. The functional unit was defined as ‘per tonne-nautical mile’ of cargo transported. A ‘well-to-wake’ system boundary was applied to each fuel pathway, covering the complete life cycle from feedstock extraction and fuel production through storage, transportation and final combustion in the ship’s engine. A detailed life cycle inventory was then developed by integrating data on energy and material flows from fuel production processes and operational profiles. To account for inherent data variability, key inventory parameters were assigned probability distributions. The static performance of all fuel pathways for a reference year was then quantified by calculating the four assessment dimensions, climate impacts, environmental sustainability, energy efficiency and economic feasibility (Climate-3E), based on the life cycle inventory. Monte Carlo simulations^{45,46} and sensitivity analysis were conducted to characterize uncertainty, yielding probability distributions for each indicator. In addition, a life cycle impact assessment^{47,48} was conducted across multiple environmental indicators to evaluate broader environmental trade-offs. Detailed inventory data sources, scenario specifications and parameter values are provided in Supplementary Information section II.

The second phase involves dynamic SD modelling, which projects how the baseline evolves in response to the interplay of policy, market and technological change. The initial input parameters for the SD model were derived from statistically representative values of the Climate-3E indicators from the static LCA. We then embedded the core mechanisms of the IMO Net-Zero Framework and the EU’s FuelEU Maritime regulation by translating key LCA outputs into dynamic variables that influence compliance costs and decision-making over time. The conceptual feedback structure was translated into a set of mathematical equations that explicitly model policy mechanisms, technology learning curves, biofuel supply constraint driven by the price elasticity, and feedback loops linking fuel economics to adoption decisions. This enabled us to simulate system behaviour quantitatively from 2024 to 2050. The validity of the SD model’s behavioural dynamics was tested by comparing its simulated output for a historical period against observed industry trends (Supplementary Fig. 7). Once validated, the model was used to project future fuel pathways, annual carbon emissions and the evolution of Climate-3E performance through 2050 under various scenarios. Detailed SD model equations, variable definitions and data sources are provided in Supplementary Information section III.

Policy scenario

This study examines three distinct regulatory scenarios to understand their individual and combined effects on the fuel transition (Supplementary Fig. 4). The first scenario models the IMO Net-Zero Framework, incorporating its ambitious targets and economic mechanisms, including increasingly strict annual limits on GHG intensity, a two-tier penalty system and its flexibility provisions. The second scenario models the EU’s FuelEU Maritime regulation, which mandates a gradual reduction in the average carbon intensity of energy used by ships calling at EU ports. This includes imposing a penalty of US\$730 per tonne of CO₂e for non-compliance and accounting for flexibility regulations, such as banking, pooling and the 50% emissions requirement for voyages involving only one EU port. The third scenario combines both frameworks, where vessels must comply with both sets of rules. In this case, more stringent requirements apply to each compliance element, and the total compliance cost is the sum of expenses incurred under each policy. This combined scenario tests the interaction and potential synergy or conflict between the global, market-based IMO policy and the regional, prescriptive EU regulation.

Assumptions and limitations

The integrated LCA–SD framework is based on several assumptions and has certain limitations. Due to the complexity of allocating shipping emissions on a global scale, this study models a round-trip voyage along the Asia–Europe Green Shipping Corridor, starting and ending at the same port. It assumes a consistent fuel blend throughout the journey, a simplification that allows clear comparison of fuels across electricity mixes and policies. For biofuels, the supply elasticity of bio-LNG is set higher than that of other biofuels, reflecting the assumption that its primary feedstocks, organic waste and residues, are widely available and can be scaled readily with demand (Supplementary Figs. 13 and 14). As the IMO Net-Zero Framework does not specify targets beyond 2040, the post-2040 ‘Base Target’ and ‘Direct Compliance Target’ values are adopted from an external source⁴⁹. The framework’s potential reward mechanism for zero-carbon fuels is not modelled, as its monetary value and operational details remain undefined, and very few fuels would qualify. The EU Emissions Trading System is not explicitly incorporated; instead, the FuelEU Maritime penalty price is used to represent the overall economic incentive from the broader EU policy mix. The model simplifies real-world operations by fixing a round-trip pattern, which may not capture all operational variability. Projections depend on technology learning curves and cost reductions, which are subject to future

uncertainty, and regional electricity mixes may evolve differently over time. Furthermore, the model does not account for ship-operator behavioural responses, future regulatory changes or macroeconomic shocks, all of which are important areas for further research.

Data availability

The life cycle inventory dataset supporting this study is available via GitHub at <https://github.com/chengjiangli/Phased-Fuel-Transitions-for-Decarbonising-the-Asia-Europe-Green-Shipping-Corridor> and via Zenodo at <https://doi.org/10.5281/zenodo.19872982> (ref. 50). All secondary data sources are described in the Analysis and its Supplementary Information. Source data are provided with this paper.

Code availability

The LCA was conducted using established software (GaBi, version 9.2.1); no custom code was generated. The SD model was built in Vensim PLE (version 10.4.0), available for unrestricted academic use at <https://vensim.com/download>.

References

- World Maritime Theme 2024: "Navigating the Future: Safety First!" (International Maritime Organization, 2023); <https://www.imo.org/en/mediacentre/pressbriefings/pages/world-maritime-theme-2024.aspx>
- Fourth Greenhouse Gas Study 2020 (International Maritime Organization, 2020); <https://www.imo.org/en/OurWork/Environment/Pages/Fourth-IMO-Greenhouse-Gas-Study-2020.aspx>
- International Shipping (International Energy Agency, 2023); <https://www.iea.org/energy-system/transport/international-shipping>
- 2018 Initial IMO Strategy (International Maritime Organization, 2018); <https://www.imo.org/en/ourwork/environment/pages/vision-and-level-of-ambition-of-the-initial-imo-strategy.aspx>
- 2023 IMO Strategy on Reduction of GHG Emissions from Ships (International Maritime Organization, 2023); <https://www.imo.org/en/ourwork/environment/pages/2023-imo-strategy-on-reduction-of-ghg-emissions-from-ships.aspx>
- Jafari, I., Soudagar, I. & Ng, S. H. *Navigating IMO Net-Zero Framework: Potential Challenges and Refinements* (Centre for Maritime Studies, National University of Singapore, 2025); <https://maritimestudies.nus.edu.sg/wp-content/uploads/2025/08/ComplianceFramework-WorkingPaper.pdf>
- More delays on Net-Zero Framework: what it means for global shipping and climate goals. *Maritime Hub* <https://maritime-hub.com/mo-delays-net-zero-framework-what-it-means-for-global-shipping-and-climate-goals> (2025).
- Decarbonising maritime transport - FuelEU Maritime. *European Commission* https://transport.ec.europa.eu/transport-modes/maritime/decarbonising-maritime-transport-fueleu-maritime_en (2024).
- Dolatabadi, S. H., Masodzadeh, P. G., Ishaq, H. & Crawford, C. Green shipping corridors: an overview of Pacific Northwest region and key ports. *Ocean Coast. Manag.* **269**, 107745 (2025).
- Pan, C. et al. Enhancing onboard carbon capture performance by tailored absorbents and process optimization: insights from pilot-scale testing and simulation. *Chem. Eng. J.* **524**, 167911 (2025).
- Mohite, A. A. & Mathew, E. Exploring the necessary upgrades in port infrastructure to accommodate and support the operation of the next-generation green ships. *Mar. Syst. Ocean Technol.* **20**, 26 (2025).
- Kersey, J., Popovich, N. D. & Phadke, A. A. Rapid battery cost declines accelerate the prospects of all-electric interregional container shipping. *Nat. Energy* **7**, 664–674 (2022).
- Wang, Y. et al. A review of low and zero carbon fuel technologies: achieving ship carbon reduction targets. *Sustain. Energy Technol. Assess.* **54**, 102762 (2022).
- Roux, M., Lodato, C., Laurent, A. & Astrup, T. F. A review of life cycle assessment studies of maritime fuels: critical insights, gaps, and recommendations. *Sustain. Prod. Consum.* **50**, 69–86 (2024).
- Jang, H., Jeong, B., Zhou, P., Ha, S. & Nam, D. Demystifying the lifecycle environmental benefits and harms of LNG as marine fuel. *Appl. Energy* **292**, 116869 (2021).
- Harahap, F., Nurdawati, A., Conti, D., Leduc, S. & Urban, F. Renewable marine fuel production for decarbonised maritime shipping: pathways, policy measures and transition dynamics. *J. Clean. Prod.* **415**, 137906 (2023).
- Helgason, R., Cook, D. & Davíðsdóttir, B. An evaluation of the cost-competitiveness of maritime fuels – a comparison of heavy fuel oil and methanol (renewable and natural gas) in Iceland. *Sustain. Prod. Consum.* **23**, 236–248 (2020).
- Ha, S., Jang, H., Park, C. & Jeong, B. A prospective life cycle assessment framework for sustainable renewable fuels in international shipping: hydrogen based e fuels. *Renew. Sustain. Energy Rev.* **226**, 116219 (2026).
- Ampah, J. D., Yusuf, A. A., Afrane, S., Jin, C. & Liu, H. Reviewing two decades of cleaner alternative marine fuels: towards IMO's decarbonization of the maritime transport sector. *J. Clean. Prod.* **320**, 128871 (2021).
- Wang, Z., Li, M., Zhao, F., Ji, Y. & Han, F. Status and prospects in technical standards of hydrogen-powered ships for advancing maritime zero-carbon transformation. *Int. J. Hydrog. Energy* **62**, 925–946 (2024).
- Larsson, A., Brackmann, C., Ragucci, R., de Joannon, M. & Fatehi, H. Numerical study of integrated biomass pyrolysis and CO₂ reforming for carbon sequestration and syngas production. *Biomass Bioenergy* **202**, 108241 (2025).
- Pavlenko, N., Comer, B., Zhou, Y., Clark, N. & Rutherford, D. *The Climate Implications of using LNG as a Marine Fuel* (International Council on Clean Transportation, 2020); https://theicct.org/sites/default/files/publications/Climate_implications_LNG_marinefuel_01282020.pdf
- Fan, G., Zhang, H., Sun, B. & Pan, F. Economic and environmental competitiveness of multiple hydrogen production pathways in China. *Nat. Commun.* **16**, 4284 (2025).
- Yan, D. et al. Carbon intensity indicator (CII) compliance: applications of ship speed optimization on each level using measurement data. *Mar. Pollut. Bull.* **212**, 117593 (2025).
- CII – Carbon Intensity Indicator. DNV <https://www.dnv.com/maritime/insights/topics/CII-carbon-intensity-indicator> (2024).
- Zamboni, G., Scamardella, F., Gualeni, P. & Canepa, E. Comparative analysis among different alternative fuels for ship propulsion in a well-to-wake perspective. *Heliyon* **10**, e26016 (2024).
- Fink, L., Esfeh, S. K., Depken, J., Ehlers, S. & Kaltschmitt, M. in *Powerfuels* (eds Bullerdiek, N., Neuling, U. & Kaltschmitt, M.) 905–940 (Springer, 2025); https://doi.org/10.1007/978-3-031-62411-7_31
- Colelli, L. et al. Liquid e-fuels for a sustainable future: a comprehensive review of production, regulation, and technological innovation. *Energy Convers. Manag.* **347**, 120529 (2026).
- Wankhede, S., Dixit, J., Sharma, D. & Sharma, N. in *Sustainable, Scalable and Storable E-Fuels for Decarbonising Transport Sector* (eds Sharma, N. et al.) 161–190 (Springer, 2025); https://doi.org/10.1007/978-981-96-7388-9_9
- Wallington, T. J. et al. Green hydrogen pathways, energy efficiencies, and intensities for ground, air, and marine transportation. *Joule* **8**, 2190–2207 (2024).

31. Nemmour, A., Inayat, A., Janajreh, I. & Ghenai, C. Green hydrogen-based E-fuels (E-methane, E-methanol, E-ammonia) to support clean energy transition: a literature review. *Int. J. Hydrog. Energy* **48**, 29011–29033 (2023).
32. Shehzad, M. F., Ishaq, H. & Crawford, C. Offshore wind energy prospects for power-to-direct air capture and power-to-gas. *Int. J. Hydrog. Energy* **48**, 39216–39224 (2023).
33. Guerra, O. J., Almajed, H. M., Smith, W. A., Somoza-Tornos, A. & Hodge, B.-M. S. Barriers and opportunities for the deployment of CO₂ electrolysis in net-zero emissions energy systems. *Joule* **7**, 1111–1133 (2023).
34. Ueckerdt, F. et al. Potential and risks of hydrogen-based e-fuels in climate change mitigation. *Nat. Clim. Change* **11**, 384–393 (2021).
35. *Renewable Capacity Statistics 2023* (International Renewable Energy Agency, 2023); <https://www.irena.org/Publications/2023/Mar/Renewable-capacity-statistics-2023>
36. Sahu, S., Kumar, P. & Dhar, A. in *Advancement in Oxygenated Fuels for Sustainable Development: Feedstocks and Precursors for Catalysts Synthesis* (eds Kumar, N. et al.) 263–283 (Elsevier, 2023); <https://doi.org/10.1016/B978-0-323-90875-7.00008-3>
37. Park, C., Jeong, B. & Zhou, P. Lifecycle energy solution of the electric propulsion ship with Live-Life cycle assessment for clean maritime economy. *Appl. Energy* **328**, 120174 (2022).
38. Jacobson, M. Z. et al. Low-cost solutions to global warming, air pollution, and energy insecurity for 145 countries. *Energy Environ. Sci.* **15**, 3343–3359 (2022).
39. Brown, T. & Hampp, J. Ultra-long-duration energy storage anywhere: methanol with carbon cycling. *Joule* **7**, 2414–2420 (2023).
40. *Draft Revised MARPOL Annex VI 2025* (International Maritime Organization, 2025); <https://wwwcdn.imo.org/localresources/en/MediaCentre/Documents/MEPC-ES.2-2-%20-%20Draft%20Revised%20MARPOL%20Annex%20VI%202025.pdf>
41. *FuelEU Maritime – Sustainable Maritime Fuels* (European Parliamentary Research Service, 2023); [https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/698808/EPRS_BRI\(2021\)698808_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/698808/EPRS_BRI(2021)698808_EN.pdf)
42. Kargbo, H., Harris, J. S. & Phan, A. N. “Drop-in” fuel production from biomass: critical review on techno-economic feasibility and sustainability. *Renew. Sustain. Energy Rev.* **135**, 110168 (2021).
43. Zoumpourlos, K., Geertsma, R., Ketterij, R. V. D. & Coraddu, A. Methanol operation in heavy-duty DICI dual-fuel engines: investigating charge cooling effects using engine combustion network spray d data. *J. Eng. Gas Turbines Power* **147**, 101007 (2025).
44. Schoenenberger, L., Schmid, A., Tanase, R., Beck, M. & Schwaninger, M. Structural analysis of system dynamics models. *Simul. Model. Pract. Theory* **110**, 102333 (2021).
45. Gholaminezhad, I., Jafarpur, K., Paydar, M. H. & Karimi, G. Multi-scale multi-objective optimization and uncertainty analysis of methane-fed solid oxide fuel cells using Monte Carlo simulations. *Energy Convers. Manag.* **153**, 175–187 (2017).
46. Ahmed, S., Li, T., Yi, P. & Chen, R. Environmental impact assessment of green ammonia-powered very large tanker ship for decarbonized future shipping operations. *Renew. Sustain. Energy Rev.* **188**, 113774 (2023).
47. Ling-Chin, J. & Roskilly, A. P. A comparative life cycle assessment of marine power systems. *Energy Convers. Manag.* **127**, 477–493 (2016).
48. Carneiro, M. L. N. M. et al. Potential of biofuels from algae: comparison with fossil fuels, ethanol and biodiesel in Europe and Brazil through life cycle assessment (LCA). *Renew. Sustain. Energy Rev.* **73**, 632–653 (2017).
49. IMO’s Net-Zero Framework and the global carbon price on shipping. *King & Spalding* <https://www.kslaw.com/news-and-insights/imos-net-zero-framework-and-the-global-carbon-price-on-shipping> (2025).
50. Li, C. chengjiangli/Phased-Fuel-Transitions-for-Decarbonising-the-Asia-Europe-Green-Shipping-Corridor (v.1.0.0). *Zenodo* <https://doi.org/10.5281/zenodo.19872982> (2026).

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C.L., X.G. and Q.Q. conceived the project and were co-lead authors. C.L., X.G., Q.Q. and W.Z. carried out analyses. C.L. and X.G. wrote the original paper. J.Y. and G.X. provided substantial scientific input in interpreting the results and editing the paper. K.F. reviewed and revised the paper. Q.Q. and W.Z. supervised the study. All authors reviewed the results and approved the final version of the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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